Beyond Co-Occurrence: Evidence-Bounded Cross-Modal Relation Ledgers for Post-Alert OT Investigation

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Abstract—Process-aware operational technology (OT) detectors identify anomalous intervals, but records grouped within an alerted event do not necessarily describe the same operational action. Temporal proximity between a network command and an electrical telemetry change does not establish execution, physical completion, or causation. We introduce the Evidence-Bounded Cross-Modal Relation Ledger (EBRL), an event-conditioned investigation method that converts network transactions and telemetry changes into fixed candidate pairs, assigns every pair a link, supported non-link, or insufficient-evidence disposition, records typed evidence paths, and limits each relation to the strongest wording warranted by the visible evidence. A deterministic publication verifier may retain, downgrade, or suppress a proposed relation but cannot access evaluator-only labels. We specify a pair-level Sherlock v3 benchmark against temporal, asset-aware, semantic-rule, direct-LLM, and evidence-ID-grounded baselines. Evaluation separates warranted-link recovery from unsupported linkage, semantic overpromotion, event-level false-link incidence, evidence-path completeness, and appropriate abstention. Reference relations are constructed automatically from evaluator-only simulator or scenario provenance where available; unresolved cases are not silently treated as negatives. [TBD: final event, episode, candidate, and reference-class counts] [TBD: principal baseline and ablation results] The study concerns evidence-bounded association within supplied event scopes, not anomaly detection, event segmentation, or unrestricted causal inference.

Index Terms—operational technology, incident investigation, cross-modal correlation, evidence grounding, large language models, abstention

I. Introduction

Process-aware OT anomaly detection has been studied through subspace, invariant, sequence, rule-based, and state-space methods [1]–[5]. Table 3 of Sherlock compares representative detectors in terms of detected attacks, false alarms, and detection time, and identifies alert actionability as an open problem [6]. These results answer whether and when a detector raises an alarm. They do not answer a different question faced after the handoff: which individual network commands and electrical/process changes may responsibly be connected in an investigation report?

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This distinction is material even when a benchmark provides complete, presegmented events. Let q_i be a command transaction and p_j a telemetry-change episode inside supplied event window W_e. In general,

$$q_i, p_j \in W_e \not\Rightarrow q_i \leftrightarrow p_j. \tag{1}$$

The records may concern different assets, incompatible operations, propagated grid effects, competing commands, or a common cause. Moreover, observing a request is not evidence of execution; observing an acknowledgement is not evidence of physical completion; and time order is not evidence of causation, compromise, intent, or attribution. A system that merges plausible observations into one narrative can therefore produce an evidentially fluent but operationally false incident story.

Prior cross-domain work correlates cyber and physical alerts [7], and recent systems use retrieval, typed claims, citations, and deterministic gates to ground incident narratives [8]–[11]. These are important precedents. The remaining methodological gap in our setting is a fixed pair-level denominator that makes unsupported links visible, distinguishes contradiction from uncertainty, and scores whether the wording of a recovered relation exceeds the evidence that supports it.

We address this gap with the Evidence-Bounded Cross-Modal Relation Ledger (EBRL). Given an externally supplied event scope, EBRL: (1) normalizes records into command and telemetry-change episodes; (2) enumerates a frozen, permissive set of candidate pairs; (3) produces one structured disposition per candidate, including a maximum relation level, typed evidence identifiers, promotion flags, and a controlled reason; (4) verifies publication boundaries using only model-visible evidence; and (5) renders the final report from admitted ledger rows. Evaluator-only provenance is isolated from both generation and publication verification and is used solely for reference construction and scoring (Fig. 1).

The study asks the following questions:

- RQ1—Relation quality: Under identical event scope, evidence budget, and candidate pairs, how does EBRL compare with temporal, asset-aware, and

semantic-rule baselines in warranted-link recovery and unsupported linkage?
- RQ2—Semantic restraint: Do relation-level boundaries, typed evidence paths, and explicit abstention improve over direct and evidence-ID-grounded LLM inference, beyond citation availability alone?
- RQ3—Method components: How does deterministic publication verification change false-link incidence, semantic overpromotion, evidence-path completeness, and warranted-link recall?

Our contributions are: (1) a pair-level EBRL method that represents links, supported non-links, and insufficient evidence without converting co-occurrence into causation; (2) an automatically scored benchmark-construction protocol that separates supported, unsupported, and indeterminate command–telemetry candidates; (3) a baseline ladder and complementary relation-quality, restraint, and event-risk metrics that cannot be satisfied by abstaining on every pair; and (4) [TBD: a concise statement of the final empirical and error-analysis contributions after rerunning all methods]. We do not claim to be the first system to use evidence grounding or deterministic verification; the contribution is their operationalization for fixed-candidate, cross-modal OT relation analysis.

II. Background and Related Work

A. Process-Aware Detection and Alert Handoff

Process-aware security monitoring uses expected dynamics rather than packet signatures alone [12]. Representative approaches include departure-based monitoring [1], learned invariants [2], sequence-to-sequence prediction [3], simple process rules [4], and comprehensible state-space models [5]. Sherlock's common evaluation is valuable upstream context [6], but its detector-level Table 3 is not a set of baselines for the present relation task: aggregate alarm outcomes do not decide which records within an alert belong to one operational action. EBRL begins after a detector or analyst has supplied a scope and therefore does not improve or evaluate detector recall, false-alarm rate, event boundaries, or alert timing.

B. Cross-Domain OT Correlation

Cross-domain alert correlation combines cyber and physical context to improve incident understanding [7]. Broader OT incident-response frameworks also coordinate analysis, recovery, and organizational decision making [13], [14]. These studies motivate temporal and asset-aware baselines, but correlation at an alert or event level is not equivalent to a command–telemetry relation with an independently fixed denominator. EBRL therefore evaluates the false links that a method proposes, not only the events or alerts it successfully associates.

C. Evidence-Grounded LLM Investigation

Retrieval-augmented LLMs have been evaluated for incident analysis [8], while Cognitive SOC couples alert enrichment with evidence-backed narrative generation [9]. Local-LLM forensic reporting is motivated by data sensitivity but continues to face citation and chronology limitations [10]. Citation correctness is also distinct from faithful reliance on a cited source [15], and abstention requires explicit evaluation rather than treating missing output as success [16]. EviGuard is a directly relevant industrial precedent for typed claims, provenance, deterministic verification, and response gating [11]. Our narrower contribution is a Sherlock-based, pair-level cross-modal task with fixed candidates, relation-level ceilings, supported non-links, and event-level false-link risk.

D. Positioning

Table I distinguishes the unit and question addressed by the closest bodies of work. It is a conceptual comparison, not a performance ranking. Detector benchmarks remain necessary upstream, whereas EBRL tests the evidentiary handoff downstream.

III. Problem Formulation

A. Event-Conditioned Scope

For event e, the evaluator supplies a window W_e and anchor t_0. Let N_e be the eligible IEC 60870-5-104 records, P_e the eligible electrical/process observations, and M_e the approved static address-to-asset and object-to-variable mappings. Retrieval produces a model-visible bundle

$$B_e = B_e^N \cup B_e^P \cup B_e^M \subseteq N_e \cup P_e \cup M_e, \tag{2}$$

where each superscript denotes the visible network, process, or mapping partition under a fixed evidence budget. Evaluator-only traces, scenario actions, attack labels, and reference relations are excluded from B_e. The conditional task is

$$(B_e, W_e, t_0) \longrightarrow (\Lambda_e, R_e), \tag{3}$$

where Λ_e is a relation ledger and R_e is a report rendered from admissible ledger entries. The task is neither anomaly detection nor event segmentation. The supplied W_e and t_0 are oracle-like inputs that permit a controlled downstream comparison while excluding detector timing error, false-alert burden, and window estimation.

B. Episodes and Frozen Candidates

Only model-visible records are normalized into network transaction episodes $Q_e = \{q_i\}$ and process-change episodes $P'_e = \{p_j\}$. A network episode groups a request, protocol response, and retransmissions without interpreting acknowledgement as execution. A process episode represents a categorical transition or sustained numerical change rather than an isolated sample. Each episode retains stable identifiers, timestamps, assets, variables, units, pre/post values, change direction, and transformation lineage. Episode metadata and the candidate manifest are declared parts of the common visible input; neither may reveal a record outside B_e.

TABLE I
Conceptual positioning relative to related OT detection and investigation work.

Work family	Unit of analysis	Primary question	Relation-evaluation distinction
Sherlock detector benchmark [6]	Event/alert	Was an attack interval detected, and when?	Upstream detector metrics; no pair denominator
Cross-domain correlation [7]	Alerts/events	Which cyber and physical alerts are correlated?	Correlation scope; no pair-level semantic ceiling
Evidence-grounded investigation [11]	Claims/reports	Is generated reasoning grounded and policy-admissible?	Claim provenance and system-specific gating
EBRL (this work)	Command–telemetry pair	What relation, if any, is warranted inside a supplied event?	Fixed pairs, explicit abstention, and pair-specific ceiling

Before any method is run, a deterministic, high-recall enumerator produces

$$\mathcal{C}_e \subseteq Q_e \times P'_e. \tag{4}$$

Candidate generation may use a development-set time bound and mappings in B_e^M, but it does not decide whether a pair is related. The same $\mathcal{C}_e$ is supplied to every method. Candidate availability and enumeration recall are measured against an evaluator-only reference universe defined in Section V so that EBRL does not receive credit for retrieval omissions or for removing difficult pairs from its denominator.

[TBD: transaction grouping, retransmission and select-before-operate handling, process change-point/deadband rule, clock-skew allowance, candidate window $\tau_{\max}$, candidate cap, deterministic sampling hash, and stratum quotas]

C. Relation Semantics

Each pair receives a disposition in $\{\mathsf{L}, \mathsf{N}, \mathsf{U}\}$: L is a supported link, N is a supported non-link backed by a visible contradiction, and U means that available evidence cannot decide. Absence of a known link is not sufficient for N. A link also receives a maximum warranted semantic level:

- R_0: no cross-modal relation is supported;
- R_1: only co-occurrence or timestamp order is supported;
- R_2: time, asset, and command–variable semantics support a bounded association;
- R_3: an independent execution or operational-response record supports a stronger relation;
- R_4: independent intervention or action–effect provenance supports physical completion or a causal statement.

Levels are ceilings on wording, not confidence scores. Execution, completion, causation, and security conclusions are additionally represented by separate Boolean promotion flags because they have different evidence requirements. In particular, a protocol acknowledgement cannot by itself set the completion flag, and an R_2 association cannot set causal or compromise flags.

IV. Evidence-Bounded Cross-Modal Relation Ledger

A. Method Overview

EBRL consists of five stages: deterministic episode normalization; frozen high-recall candidate construction; structured relation disposition; visible-evidence boundary verification; and controlled report synthesis. Figure 1 shows the evaluated path and the separate evaluator-only path. The method's central object is the ledger rather than the final prose: every candidate must be accounted for, uncertainty remains visible, and report statements can be traced back to a relation decision.

B. Episode Construction

Normalization applies frozen transformations

$$Q_e = g_N(B_e^N; \theta_N), \qquad P'_e = g_P(B_e^P; \theta_P). \tag{5}$$

The network function groups packets using flow, ASDU type, common address, information-object address, cause of transmission, and a bounded transaction interval. The process function identifies a state transition or change segment and records its precondition, postcondition, magnitude, direction, and interval. Neither function receives attack labels or reference links. The resulting episode IDs, source-record IDs, and transformations are included in the case trace.

[TBD: complete values for θ_N and θ_P, duplicate policy, numerical tolerances, and worked normalization examples]

C. Permissive Candidate Construction

A preliminary candidate rule is

$$\mathcal{C}_e = \{(q_i, p_j) : |\Delta t_{ij}| \le \tau_{\max} \ \vee \ \mu(a(q_i), a(p_j)) = 1\}, \tag{6}$$

where $\Delta t_{ij} = t(p_j) - t(q_i)$ and μ is a frozen asset-neighborhood mapping. Candidates are constructed over episodes rather than a raw-record Cartesian product. The retained strata include mapped-compatible pairs, mapped but semantically incompatible pairs, near-time wrong-asset pairs, reverse-time pairs, and competing-command pairs. If a cap is necessary, selection is performed by frozen stratum quotas and a deterministic hash. Candidate construction is intentionally permissive: asset, timing, and semantic tests that decide a final link are left to the evaluated methods.

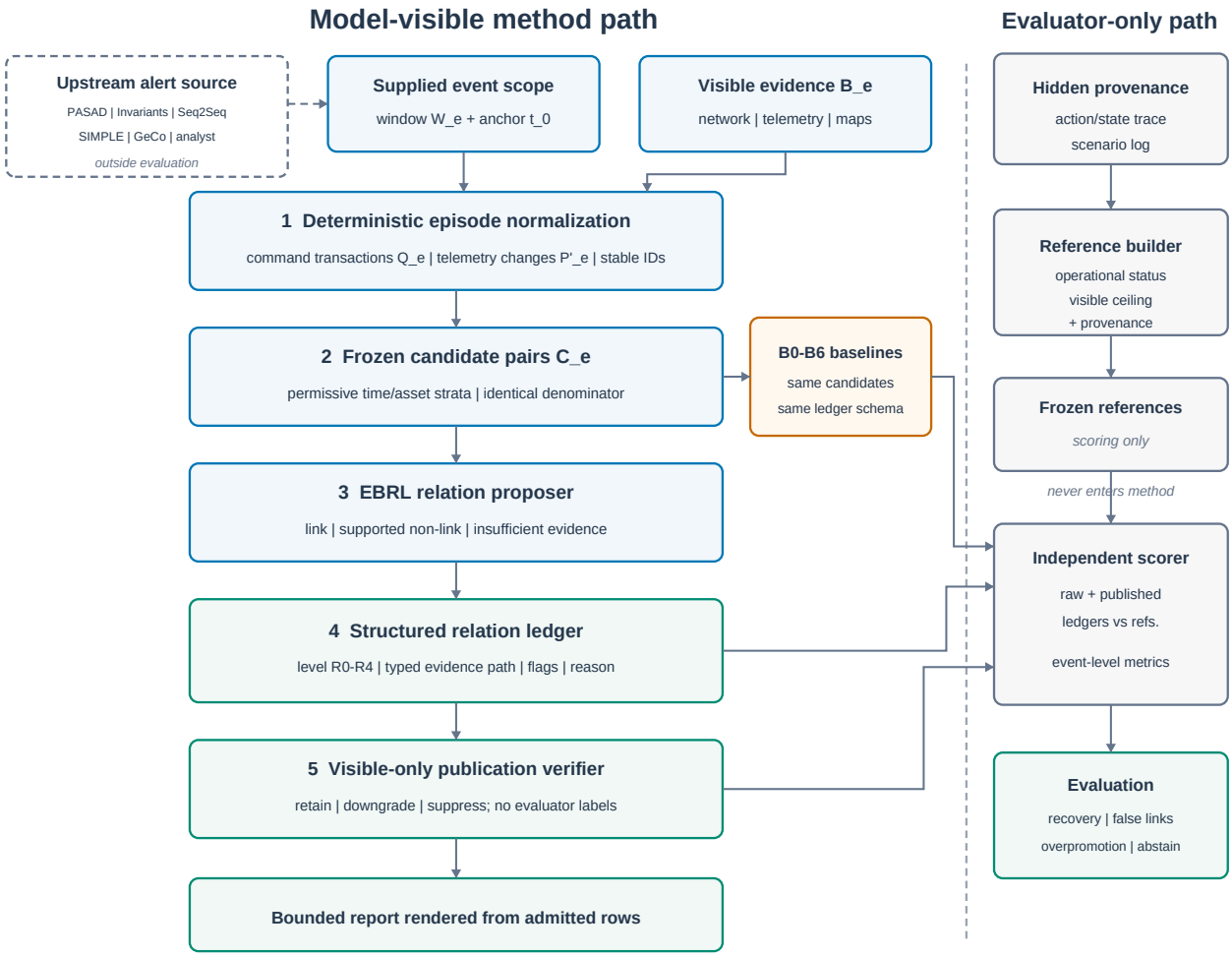


Fig. 1. EBRL workflow. An upstream detector such as those evaluated in Sherlock's Table 3 [1]–[5], or an analyst, supplies the event scope but is outside the evaluation. Every method receives the same visible evidence and frozen candidate pairs. The publication verifier has no access to evaluator-only provenance; the reference builder is used only by the scorer.

D. Ledger Schema and Decision Policy

For each candidate $c = (q_i, p_j)$, EBRL emits

$$\lambda_c = (i, j, \hat{y}_c, \hat{\lambda}_c, \hat{\mathbf{z}}_c, \hat{s}_c, S_c^N, S_c^P, S_c^M, \rho_c), \tag{7}$$

where $\hat{y}_c \in \{\mathsf{L}, \mathsf{N}, \mathsf{U}\}$ is the disposition; $\hat{\lambda}_c \in \{0, \ldots, 4\}$ is the proposed semantic level; $\hat{\mathbf{z}}_c$ contains execution, completion, causal, and security promotion flags; $\hat{s}_c$ is an optional confidence; S_c^N, S_c^P, S_c^M are typed network, process, and mapping evidence sets; and ρ_c is a controlled reason code. The parser also records a validity bit h_c. A missing or malformed row is $h_c = 0$, receives no link, non-link, or abstention credit, and is reported as a ledger-completeness failure; it is not silently converted to U.

The relation proposer evaluates temporal order, asset compatibility, command–variable semantics, change direction or target, and competing explanations. It uses N only when a visible contradiction supports rejection; otherwise an unsupported candidate receives U. A proposed R_2 link must cite both modalities and the applicable mapping. Higher levels additionally require independent operational evidence. The implementation may be rule based, LLM based, or hybrid, but it must emit the same complete ledger schema.

[TBD: the EBRL proposer prompt or rule set, controlled reason vocabulary, semantic compatibility ontology, confidence production, and pseudocode]

E. Publication Verifier and Scorer Separation

The visible-only publication verifier V_{pub} checks schema validity, candidate membership, evidence visibility, units, timestamps, mapping support, semantic compatibility, acknowledgement overpromotion, and level-specific evidence requirements. It may retain a row, downgrade its level, or rewrite its published disposition to U with a verifier reason. It never deletes a candidate row and preserves the raw disposition in the audit trace. It cannot add evaluator evidence, infer a hidden reference link, or convert uncertainty into a supported non-link.

TABLE II
Required EBRL ledger fields and their evaluation role.

Field	Meaning	Boundary requirement
Episode IDs	Frozen command–telemetry candidate	Both IDs must exist in $\mathcal{C}_e$
Row validity	Complete, unique, schema-valid candidate row	Invalid or missing rows receive no disposition credit
Disposition	L, N, or U	Non-link requires contradiction; missing evidence requires U
Relation level	Maximum proposed wording, R_0–R_4	Must not exceed the visible evidence ceiling
Promotion flags	Execution, completion, causal, and security assertions	Each flag has an independent evidence predicate
Typed evidence sets	Network, process, and mapping identifiers	IDs must be visible and assigned to valid roles
Reason code	Sufficient association, mapping/time/semantic contradiction, competition, or insufficiency	Must agree with the disposition
Confidence	Optional selective-publication score	Used only if prespecified and calibrated on development data

A separate scorer S_{eval} compares raw and published ledgers with frozen references. The reference builder, V_{pub}, and S_{eval} are implemented and versioned separately. Metrics derived from the same visible predicates used by V_{pub} are labeled policy-consistency metrics. Primary claims of relation accuracy require evaluator-only operational provenance or counterfactual labels not available to V_{pub}. This design directly addresses verifier–scorer circularity: a retained-set value of 1.0 cannot be interpreted as independent factual validation merely because the same policy filtered and scored the output.

[TBD: the complete verifier rule table, downgrade precedence, treatment of multiple failures, and independent unit/challenge tests]

F. Controlled Report Synthesis

The final report is rendered only from admitted ledger rows using controlled templates. It presents supported relations, supported non-links when operationally relevant, and unresolved candidates or missing evidence. Free text is not permitted to introduce additional cross-modal relations. This makes ledger-to-report consistency mechanically checkable and avoids asking experts to grade every generated narrative. If a future condition permits free-form prose, semantic agreement with the ledger must be evaluated separately.

G. Complexity and Reproducibility

Episode formation is linear in the visible records. A naive candidate enumerator is $O(|Q_e||P'_e|)$, while time indexing and asset partitions reduce the number of evaluated pairs in practice. The complete experiment records versions of the episode builder, mapping, candidate manifest, proposer, verifier, templates, model configuration, and scorer.

[TBD: candidate-builder complexity after indexing, observed candidates per event, wall-clock time, token consumption, and artifact hashes]

V. Benchmark and Automatic Reference Construction

A. Dataset and Split

The study uses Sherlock v3, a Wattson power-grid co-simulation dataset containing IEC 60870-5-104 traffic and electrical/process observations [6], [17]. The current split contains 16 development events from 01-Basic and 32 evaluation events: 18 Semiurban and 14 Rural events, balanced across 16 cyber and 16 benign cases. The dataset and these events were previously inspected; the revised study is therefore a constructed benchmark, not an untouched confirmation sample or an estimate of operational prevalence. All EBRL outputs, automatic references, baselines, and scores will be regenerated under the present pair-level protocol. Development events are used to freeze thresholds, ontology entries, prompts, and candidate rules; evaluation events are not used for tuning.

[TBD: confirm the final event manifest and report per-scenario episode and candidate counts]

B. Evaluator-Only Relation References

The reference contract deliberately has two axes. For each candidate c, the builder produces

$$r_c = (o_c^*, g_c, v_c^*, \lambda_c^{\text{vis}}, \mathbf{z}_c^{\text{vis}}, u_c). \tag{8}$$

The evaluator-only operational status $o_c^* \in \{\mathsf{L}, \mathsf{N}\}$ states whether the pair is operationally linked or unlinked, and $g_c \in \{0, 1\}$ states whether that status is independently evaluable. When $g_c = 0$, o_c^* is undefined and the pair is excluded from factual relation metrics. Separately, the visible-evidence disposition $v_c^* \in \{\mathsf{L}, \mathsf{N}, \mathsf{U}\}$ states whether B_e warrants a link, warrants a non-link through contradiction, or is insufficient. The ceiling λ_c^{vis} and flags $\mathbf{z}_c^{\text{vis}}$ define the strongest admissible wording. Finally, u_c records whether the source is simulator-grounded, observational, counterfactual-by-construction, rule-derived, or unresolved. Thus, an unknown operational truth is not mislabeled as a correct abstention, and visible insufficiency is not mislabeled as an operational non-link.

The preferred source for o_c^* is an evaluator-only action identifier, simulator control/action trace, scenario operation log, or direct state-transition provenance not included

in B_e. A natural positive requires an independent join from the operation to the response. A natural negative requires a trace that is demonstrably complete enough to rule out the candidate relation; absence from an incomplete positive list is not negative evidence. Approved visible mappings, timing, semantics, and change direction determine v_c^* and the wording ceiling. Exact asset, type, direction, or ordering contradictions may support $v_c^* = \mathsf{N}$, while competing commands, incomplete mappings, indirect propagation, and ambiguous alignment produce $v_c^* = \mathsf{U}$.

Visible time, mapping, and semantic rules can establish an R_2 evidence ceiling, but they are not an independent operational reference if a baseline or verifier uses the same rules. If Sherlock exposes no independent command-to-action or action-to-effect provenance, $g_c = 0$ for the affected natural pairs, operational precision/recall are reported as NA, and conclusions are limited to agreement with a rule-derived evidence-admissibility contract. R_3 and R_4 factual accuracy will then be excluded rather than inferred from co-occurrence.

Let $\mathcal{U}_e^{\mathrm{op}}$ be the evaluator-only universe of all independently provenance-linked command–effect pairs inside W_e, constructed before applying the evidence budget, and let $\mathcal{U}_e^{\mathrm{vis}}$ be the subset whose two episodes are representable from B_e. Evidence-availability recall is $|\mathcal{U}_e^{\mathrm{vis}}|/|\mathcal{U}_e^{\mathrm{op}}|$, while candidate-enumeration recall is $|\mathcal{C}_e \cap \mathcal{U}_e^{\mathrm{vis}}|/|\mathcal{U}_e^{\mathrm{vis}}|$. These diagnostics distinguish retrieval loss from candidate-builder loss. They are defined only if the operational universe is independently observable.

[TBD: identify the exact Sherlock fields, completeness assumptions, and joins used for positive and negative operational provenance; define $\mathcal{U}_e^{\mathrm{op}}$ and episode matching; otherwise freeze the study at R_0–R_2 and characterize it as policy/silver-label evaluation]

C. Hard-Negative Challenge Set

Natural evaluation is supplemented, but not mixed, with matched challenge pairs. A deterministic transformation retains the original event and lag stratum while swapping the asset, changing command semantics or direction, reversing admissible order, pairing an acknowledgement without compatible telemetry, or crossing competing commands. These cases test whether a method resists plausible but incorrect links. Constructed negatives are reported separately because their designed frequency does not estimate natural false-link prevalence.

[TBD: counterfactual operators, validity checks, matching variables, number per stratum, random seed, and leakage audit]

D. Human Validation Policy

Primary scoring is designed to be automatic and does not require experts to review every report. Human labeling is unnecessary only if independent provenance supports the operational reference and the final narrative is a deterministic rendering of the ledger. If provenance is incomplete, expert review becomes necessary for reference validity, not for repeatedly grading each method's prose. In that contingency, two OT reviewers will independently and blindly audit a prespecified stratified sample of supported links, supported non-links, unresolved cases, hard negatives, builder conflicts, and all proposed R_3/R_4 references. Disagreements will be adjudicated and class-specific agreement will be reported. A sampled audit estimates reference error but does not turn unreviewed silver labels into complete factual gold; factual metrics require either independent provenance or expert labeling of the entire evaluable set.

[TBD: choose one path: document why a fully automatic reference is sufficient, or specify reviewer identities/qualifications, sample size, sampling seed, blinding, adjudication, agreement statistic, and label-change rate]

If no independent expert audit is performed, all conclusions will be described as conditional on the frozen automatic reference contract rather than domain-validated factual correctness or analyst utility.

TABLE III
Planned automatic reference sources and the claims they permit.

Reference source	Automatically evaluated property	Limitation
Visible timestamps, mappings, and semantics	v_c^*, λ_c^{vis}, and permitted wording	Policy consistency, not operational truth
Evaluator-only action/state provenance	o_c^* and g_c for operational links and, only if complete, non-links	[TBD: confirm availability, completeness, and independence]
Matched counterfactual construction	Operational non-link by construction plus visible contradiction class	Robustness challenge, not natural prevalence
Gold-unresolved natural candidate	$g_c = 0$; no factual score	May still support visible-policy evaluation

VI. Baselines and Experimental Conditions

A. Baseline Ladder

Table IV defines the planned comparison. Every method receives the same B_e, $\mathcal{C}_e$, stable identifiers, and mappings, and emits exactly one row per candidate in the same ledger schema. An unmatched candidate defaults to U; it becomes N only when the method identifies a frozen, visible contradiction. Rule-based methods attach evidence IDs mechanically and use the same report templates. Thresholds, compatibility tables, and learned-model hyperparameters are selected only on development events and frozen before final evaluation.

B2 and B3 are independent baselines only when their visible decision rules are not also treated as operational ground truth. If the automatic reference is built from the same asset/time/semantic predicates, these conditions will be labeled reference-rule replay diagnostics or policy ceilings rather than independent evidence of accuracy. B4 is included only if the development split contains enough independently labeled pairs to train and choose a threshold without evaluation leakage; otherwise its infeasibility and class counts are reported. A full-trace or exact-mapping oracle, if reported, is likewise an upper-bound diagnostic and never a deployable baseline.

B. Input–Output Parity

The principal comparison uses the combined EN evidence view because a cross-modal relation cannot be assessed from one modality alone. Every method receives the same supplied W_e and t_0, a fixed-budget B_e^{EN}, the same candidate list, and the same approved mapping. No method receives reference dispositions, full-resolution evaluator traces, attack labels, or scenario descriptions. The earlier electrical-only and network-only conditions are retained only as missing-modality controls, not as primary relation methods.

The deterministic baselines and B4 are run once after freezing. Generative conditions use the same model snapshot, decoding parameters, token limit, and planned repetition count. All conditions contain the same output fields, although B5 does not require nonempty evidence sets. This permits operational pair identity to be scored independently from grounding. Missing candidate rows, invalid identifiers, and malformed outputs remain $h_c = 0$ failures rather than being repaired or credited as abstentions. Candidate availability and enumeration recall are reported separately from relation-decision quality.

[TBD: combined-view record or token budget, model snapshot, prompts, decoding parameters, schema, repetition count, threshold-selection procedure, run dates, and missing-run policy]

TABLE IV
Cross-modal relation baselines and EBRL conditions.

ID	Method	Relation rule	Purpose
B0	Abstain-all	Outputs U for every candidate	Zero-link-risk reference; exposes the need for warranted-link recall
B1	Nearest-after	Links the nearest subsequent change within frozen τ and assigns the honest R_1 ceiling	Tests pair selection from temporal proximity alone
B2	Asset+Time	Links a mapped asset inside the time window; R_1 unless semantics independently support R_2	Conventional cross-domain correlation baseline
B3	Asset+Time+Semantic	Proposes an R_2 link after adding command–variable type, target/direction, change threshold, and competition logic	Strongest deterministic non-LLM baseline
B4	Shallow pair classifier	Development-trained logistic/ranking model over time, asset, type, direction, magnitude, and competition features	Learned non-LLM baseline, if development labels suffice
B5	Direct LLM	Receives the candidates and output fields but no evidence-boundary policy or mandatory evidence path	Tests unconstrained relation inference
B6	ID-grounded LLM	Requires visible evidence IDs but omits level-specific evidence predicates and the deterministic verifier	Tests whether citation alone prevents false linkage
P0	EBRL without verifier	Uses the EBRL proposer and complete raw ledger but bypasses V_{pub}	Component ablation
P1	Full EBRL	Applies exact deterministic V_{pub} replay to P0 while preserving every row	Proposed complete method

VII. Evaluation Metrics and Statistical Analysis

A. Automatic Relation Metrics

The primary factual metrics use only candidates with independent operational labels ($g_c = 1$). Let $p_c = \mathbb{1}[h_c = 1 \wedge \hat{y}_c = \mathsf{L}]$. Operational Link Precision (OLP), Operational Link Recall (OLR), and Unsupported Operational Link Rate (UOLR) are

$$\mathrm{OLP} = \frac{\sum_c g_c p_c \mathbb{1}[o_c^* = \mathsf{L}]}{\sum_c g_c p_c}, \tag{9}$$

$$\mathrm{OLR} = \frac{\sum_c g_c p_c \mathbb{1}[o_c^* = \mathsf{L}]}{\sum_c g_c \mathbb{1}[o_c^* = \mathsf{L}]}, \tag{10}$$

$$\mathrm{UOLR} = \frac{\sum_c g_c p_c \mathbb{1}[o_c^* = \mathsf{N}]}{\sum_c g_c \mathbb{1}[o_c^* = \mathsf{N}]}. \tag{11}$$

These metrics evaluate pair identity independently of citations, semantic ceilings, and verifier predicates. OLR prevents abstain-all from appearing successful. If the dataset cannot support $g_c = 1$ labels, all three are NA rather than replaced by visible-rule agreement.

Evidence admissibility is evaluated on the separate visible axis. Let $\mathrm{PathOK}(c)$ indicate that cited identifiers exist in B_e, have valid roles, and form the path required by the proposed level, and define

$$b_c = \mathbb{1}[h_c = 1, \hat{y}_c = \mathsf{L}, v_c^* = \mathsf{L}, \hat{\lambda}_c \le \lambda_c^{\mathrm{vis}}, \hat{\mathbf{z}}_c \preceq \mathbf{z}_c^{\mathrm{vis}}, \mathrm{PathOK}(c)]. \tag{12}$$

Evidence-Admissible Link Precision (EAP) and Visible-Warranted Link Recall (V-WLR) are

$$\mathrm{EAP} = \frac{\sum_c b_c}{\sum_c \mathbb{1}[h_c = 1, \hat{y}_c = \mathsf{L}]}, \tag{13}$$

$$\text{V-WLR} = \frac{\sum_c b_c}{\sum_c \mathbb{1}[v_c^* = \mathsf{L}]}. \tag{14}$$

These are explicitly policy/admissibility metrics. A post-verifier increase demonstrates enforcement consistency, not independent factual improvement.

Supported Non-Link Recall (NLR) and Visible-Indeterminacy Abstention Recall (IAR) keep contradiction separate from insufficient evidence and give no credit to invalid rows:

$$\mathrm{NLR} = \frac{\sum_c \mathbb{1}[h_c = 1, v_c^* = \mathsf{N}, \hat{y}_c = \mathsf{N}]}{\sum_c \mathbb{1}[v_c^* = \mathsf{N}]}, \tag{15}$$

$$\mathrm{IAR} = \frac{\sum_c \mathbb{1}[h_c = 1, v_c^* = \mathsf{U}, \hat{y}_c = \mathsf{U}]}{\sum_c \mathbb{1}[v_c^* = \mathsf{U}]}. \tag{16}$$

Semantic Overpromotion Rate (SOR) and Evidence-Path Completeness (EPC) are

$$\mathrm{SOR} = \frac{\sum_c \mathbb{1}[h_c = 1, \hat{y}_c = \mathsf{L}, v_c^* = \mathsf{L}, (\hat{\lambda}_c > \lambda_c^{\mathrm{vis}} \vee \hat{\mathbf{z}}_c \not\preceq \mathbf{z}_c^{\mathrm{vis}})]}{\sum_c \mathbb{1}[h_c = 1, \hat{y}_c = \mathsf{L}, v_c^* = \mathsf{L}]}, \tag{17}$$

$$\mathrm{EPC} = \frac{\sum_c \mathbb{1}[h_c = 1, \hat{y}_c = \mathsf{L}, \mathrm{PathOK}(c)]}{\sum_c \mathbb{1}[h_c = 1, \hat{y}_c = \mathsf{L}]}. \tag{18}$$

A severity companion averages $(\hat{\lambda}_c - \lambda_c^{\mathrm{vis}})_+$ over asserted links. EPC is structural provenance, not relation truth. Every ratio is NA when its denominator is zero.

One operationally false relation can contaminate an otherwise useful report. Event False-Relation Incidence (EFRI) therefore uses only independent non-links:

$$\mathrm{EFRI} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \mathbb{1}[\exists c \in \mathcal{C}_e : g_c = 1 \wedge o_c^* = \mathsf{N} \wedge p_c = 1]. \tag{19}$$

An analogous Event Policy-Violation Incidence reports whether any asserted row fails b_c, but is labeled policy consistency rather than false relation. Operational-label coverage, valid-row coverage, pair-assertion coverage, event-assertion coverage, evidence-availability recall, and

candidate-enumeration recall are mandatory diagnostics. If a prespecified confidence is available, risk–coverage and area under the risk–coverage curve are secondary measures. Overall pair accuracy and a single composite score are not primary because class imbalance and abstention can make them misleading.

B. Aggregation and Inference

For generative methods, numerator and denominator sufficient statistics are first averaged across valid repetitions within each event. Primary ratios then pool those event contributions across the complete evaluation set; a method cannot remove a difficult event by producing no links. Assertion and valid-row coverage are reported beside every precision result. Event-macro values under a predeclared zero-denominator policy are reported as sensitivity analyses rather than replacing the pooled estimand.

Uncertainty is summarized with a scenario- and attack-family-stratified bootstrap that resamples events and recomputes each pooled ratio. Candidate pairs and repetitions are never treated as independent samples. Paired contrasts reuse the same resampled events for both methods rather than dropping events with undefined per-event precision. The primary predeclared contrasts are P1–B3, P1–B6, and P1–P0. The first tests full EBRL beyond deterministic semantic correlation, the second tests relation boundaries beyond citation, and the third isolates publication-verifier effects.

[TBD: bootstrap resamples and seed, confidence level, multiplicity policy, minimum defined-event count, and handling of malformed/provider-incomplete runs]

VIII. Results

At this revision stage, the pair-level reference constructor, baseline reruns, and EBRL ablations are not complete. The earlier claim-support and protocol-exact reconstruction measurements are intentionally not carried into this section because they answer a different task and do not provide a valid relation-level baseline. No numerical EBRL performance claim is made until the frozen candidate manifest, independent reference path, and scorer are complete.

A. RQ1—Relation Quality

[TBD: report P1 against B1–B4 on OLP, OLR, UOLR, and EFRI; distinguish natural candidates from constructed hard negatives; include paired event-bootstrap contrasts and coverage]

B. RQ2—Semantic Restraint

[TBD: report P1 against B5 and B6 on factual relation metrics where available and on EAP, V-WLR, SOR, IAR, NLR, and promotion-flag errors; determine whether evidence IDs alone prevent unsupported linkage]

C. RQ3—Verifier Contribution and Ablations

[TBD: report P1–P0 factual relation changes separately from mechanical policy-compliance changes, then ablate three-way disposition, semantic ceilings, role-typed evidence paths, competition handling, and publication verification]

D. Error and Qualitative Analysis

[TBD: counts and representative cases for wrong asset, wrong order, incompatible semantics, acknowledgement-to-completion promotion, competing commands, indirect propagation, missing mapping, and malformed evidence paths]

The qualitative appendix will include one correctly supported link, one correctly unresolved pair, and one false link from the strongest baseline. Each case will show the two episodes, visible evidence path, reference source, predicted and maximum levels, and the reason for the disposition. Examples will be selected by a frozen rule rather than only for rhetorical convenience.

TABLE V
Composition of the frozen pair-level benchmark.

Split	Events	Net. ep.	Proc. ep.	Candidates	Op. L	Op. N	Op. unresolved	Vis. U	Hard neg.
Development	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Evaluation	32	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

TABLE VI
Planned factual pair-relation results using only independently evaluable operational references. All values are NA if such provenance is unavailable.

Method	OLP↑	OLR↑	UOLR↓	EFRI↓	Link coverage
B0 Abstain-all	NA	TBD	TBD	TBD	TBD
B1 Nearest-after	TBD	TBD	TBD	TBD	TBD
B2 Asset+Time	TBD	TBD	TBD	TBD	TBD
B3 Asset+Time+Semantic	TBD	TBD	TBD	TBD	TBD
B4 Shallow classifier	TBD	TBD	TBD	TBD	TBD
B5 Direct LLM	TBD	TBD	TBD	TBD	TBD
B6 ID-grounded LLM	TBD	TBD	TBD	TBD	TBD
P0 EBRL w/o verifier	TBD	TBD	TBD	TBD	TBD
P1 Full EBRL	TBD	TBD	TBD	TBD	TBD

TABLE VII
Planned visible-evidence policy and abstention results. These scores measure admissibility under the frozen contract, not independent operational truth.

Method	EAP↑	V-WLR↑	NLR↑	IAR↑	SOR↓	EPC↑	Valid rows↑
B0 Abstain-all	NA	TBD	TBD	TBD	NA	NA	TBD
B1 Nearest-after	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B2 Asset+Time	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B3 Asset+Time+Semantic	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B4 Shallow classifier	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B5 Direct LLM	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B6 ID-grounded LLM	TBD	TBD	TBD	TBD	TBD	TBD	TBD
P0 EBRL w/o verifier	TBD	TBD	TBD	TBD	TBD	TBD	TBD
P1 Full EBRL	TBD	TBD	TBD	TBD	TBD	TBD	TBD

TABLE VIII
Planned component ablations.

Condition	OLP	OLR	EAP	SOR	EFRI
Full EBRL (P1)	TBD	TBD	TBD	TBD	TBD
No publication verifier (P0)	TBD	TBD	TBD	TBD	TBD
Binary link/non-link only	TBD	TBD	TBD	TBD	TBD
No semantic ceiling	TBD	TBD	TBD	TBD	TBD
Untyped evidence IDs	TBD	TBD	TBD	TBD	TBD

IX. Discussion

A. What the Evaluation Can Establish

The proposed evaluation separates properties that are often collapsed in post-alert reporting. OLP, OLR, UOLR, and EFRI use independent operational labels to evaluate pair identity. EAP, V-WLR, NLR, IAR, and SOR instead evaluate compliance with the visible evidence contract. EPC verifies trace structure but cannot establish factual truth. This separation prevents a verifier's own policy from mechanically becoming evidence that the recovered relation is operationally correct.

The three-way disposition is central. Treating every unlisted pair as a negative would reward unsupported certainty and conceal ambiguity. Treating every silence or malformed row as abstention would let incomplete outputs appear safe. Requiring one valid row per frozen candidate makes both failure modes observable. Event-level EFRI further reflects the operational concern that a single false relation may distort an otherwise accurate timeline.

B. Automation and the Remaining Role of Experts

Automatic scoring is preferable when simulator or scenario provenance is independently available, because every method can be compared repeatedly without subjective report grading. Expert labor should then be used for a stratified validity audit rather than as the primary scorer. If independent provenance is absent, however, removing experts entirely would merely replace visible human judgment with an unvalidated rule. The defensible fallback is a one-time, blinded, dual-reviewer audit or pair annotation, with causal claims removed unless reviewers have evidence beyond timing and mapping.

C. Deployment Interpretation: Skills and Boundary Safety

EBRL can be implemented as a versioned investigation skill: a procedure contract defining required inputs, permitted evidence sources, candidate and output schemas, stop conditions, and review state. The skill does not itself grant data access. A production deployment would require

case-scoped read-only tools, identity and authorization controls, data-residency and egress policy, and tamper-resistant trace storage, consistent with zero-trust principles [18] and OT guidance for secure AI integration [19]. The present public-data experiment evaluates none of these enforcement properties. It evaluates only whether bounded inputs produce a reproducible relation ledger and whether the publication boundary is followed.

X. Threats to Validity

The supplied event window and anchor are oracle-like; EBRL is not exposed to false alerts, boundary error, or detector latency. Candidate-generator failures cap relation recall, while an overly broad generator changes class balance and cost. The event set comes from one co-simulation family and was previously inspected, so it is neither a natural prevalence sample nor untouched confirmatory evidence.

Reference validity is the largest dependency. Simulator traces may record software actions without proving real physical causation. Conversely, a gold label reconstructed from the same asset/time/semantic rules as B3 or the verifier creates circularity and must be described as policy agreement. Missing mappings, clock misalignment, indirect propagation, and competing commands may expand the unresolved class. Constructed negatives test controlled robustness but do not estimate their frequency in deployment.

Pairs within an event and repeated model outputs are dependent, which is why inference is event-level. Threshold tuning, ontology construction, prompt revision, and candidate filtering can leak evaluation information unless frozen on development events. Model and provider variation, class imbalance, malformed outputs, shared packet ancestry across modalities, and incomplete runs may affect comparisons. Finally, deterministic report rendering establishes ledger consistency but not analyst usefulness; free-form prose, local deployment, security enforcement, and human decision benefit require separate studies.

XI. Conclusion

This work reframes post-alert OT investigation as fixed-candidate, evidence-bounded cross-modal relation analysis. EBRL requires every command–telemetry candidate to receive a link, supported non-link, or insufficient-evidence disposition together with a semantic ceiling, typed evidence path, and reason. It separates visible-only publication verification from evaluator-only reference construction and evaluates both recovery and restraint against temporal, asset-aware, semantic, and LLM baselines. [TBD: insert one sentence summarizing the principal numerical result after the revised experiment is complete] The resulting claims will remain conditional on supplied event scopes and the validated reference source; they will not be presented as end-to-end anomaly detection, unrestricted causal inference, or production analyst utility.

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